Carbon-Aware Scheduling for Phase-Disaggregated LLM Inference in Cloud and Serverless Platforms: Review and Simulation

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***Abstract*—Interactive large language model (LLM) inference separates into compute-bound prefill and memory-bandwidth-bound decode stages across multi-region cloud clusters and bursty serverless GPU pools. A PRISMA-2020-guided review of 65 works shows that while recent systems address subsets in isolation (including single-cluster disaggregated GreenLLM and colocated multi-datacenter carbon routers), no implemented system jointly integrates prefill/decode disaggregation, wide-area key-value (KV) transfer gating, host and GPU embodied carbon amortization, and serverless checkpoint keep-alive, and 76.9% of reviewed studies omit manufacturing emissions. To evaluate this design space, I formalize an ISO/IEC 21031:2024 Software Carbon Intensity (SCI) control architecture in a trace-driven SimPy 4.1.2 discrete-event simulator validated against published serving benchmarks and held-out scaling trends. Over 10 paired seeds driven by Azure LLM traces, UK NESO, and Japan TSO grid telemetry, hybrid routing lowers operational SCI by up to 60.8% and total SCI by up to 54.9% on conversational traffic at 0.81%–0.82% SLO violations, prunes cross-region KV pipelining below 0.31% even on 400 Gbps links, and reduces bursty-code SLO violations by 85.2%–86.0% below unconditional geographic disaggregation. On wind-clean grids, amortized hardware reaches 93.0%–97.6% of total SCI, causing four spare clean nodes to raise lifecycle emissions by 27.7%–32.6% at nominal load before turning carbon-reducing (-30.6% to -36.5% at 3.0x) once demand crosses a 1.6x–1.7x break-even threshold.**

***Index Terms*—Carbon-aware scheduling, embodied carbon, large language model inference, phase disaggregation, serverless GPUs, Software Carbon Intensity.**

I. Introduction

LARGE language model (LLM) inference has shifted cloud datacenter energy demand from offline batch training toward continuous, latency-critical serving. Survey estimates indicate that U.S. datacenter electricity consumption could climb from 176 TW h in 2023 toward 580 TW h by 2028 [1]. Because interactive LLM serving runs across distributed cloud clusters, its net carbon footprint depends on compute power draw, temporal and regional electricity grid carbon intensity (gCO_2eq/kWh), and hardware manufacturing emissions amortized over server and accelerator lifetimes [2], [3].

Two distinct research communities have tackled pieces of this problem over the past five years, yet their mechanisms remain only partially connected. First, systems researchers have shown that autoregressive LLM generation divides into a compute-bound *prefill* phase (governing time-to-first-token, TTFT) and a memory-bandwidth-bound *decode* phase (governing time-per-output-token, TPOT) [4]–[6]. Disaggregating prefill and decode across separate GPU pools eliminates phase interference and enables power capping or lower-tier decode GPUs [4], [7]–[9]. Second, sustainable cloud researchers have built carbon-aware spatial/temporal routers and serverless keep-alive controllers that follow low-carbon regions or clean grid hours [10]–[14].

A PRISMA-2020-guided literature review of 65 retained works (51 empirical/modeling studies, 11 surveys or agendas, 2 standards, and 1 production trace release) clarifies both recent 2024–2026 progress and the remaining architectural gap. On single clusters, GreenLLM [15] pairs heterogeneous P/D disaggregation with grid CI and GPU lifetime reuse, while SPROUT [16] and GreenCache [17] model operational plus hardware/SSD embodied carbon. Across multiple regions, recent geo-distributed LLM routers [18]–[24] cut operational emissions over colocated pools. Yet across all 65 works, 50/65 (76.9%) omit manufacturing emissions, 0/11 implemented cross-region carbon schedulers evaluate separated P/D pools with wide-area KV transfer gating (FREESH [22] uses P/D notation in Eq. 1c while evaluating colocated pools, and CarbonRouter [20] defers WAN KV migration), InFactPlanner [25] abstracts away batching, KV state, and P/D separation, and Sisodia's `llm-d` agenda [26] has zero experiments. Thus, no implemented system jointly integrates P/D disaggregation, cross-region carbon routing with WAN KV gating, host plus GPU embodied accounting, and serverless keep-alive across 10.3–84.0 s 70B loader tiers [27].

To evaluate this joint intersection under controlled, reproducible conditions, I conduct a PRISMA-2020-guided review paired with a trace-driven discrete-event simulation study structured around four research questions:

- **RQ1 (Taxonomy and Joint-Design Gaps):** Where do existing carbon-aware cloud schedulers, geo-distributed LLM routers, serverless keep-alive controllers, and phase-disaggregated serving engines intersect, and what remains unimplemented at their boundary?
- **RQ2 (Disaggregated Carbon–SLO Frontier):** Under real UK/Japan grid telemetry and production Azure LLM traces, when does hybrid carbon-and-queue-aware phase routing outperform latency-only colocation and phase-agnostic or unconditional spatial routing under TTFT/TPOT SLOs?
- **RQ3 (Serverless Keep-Alive Economics):** Across carbon-intensive and wind-clean grids, how does hazard-driven dynamic keep-alive trade operational carbon against cold-

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start SLO violations as loader latency spans fast NVMe streaming to standard deserialization?

- **RQ4 (Embodied Scale-Out Paradox):** When does adding spare GPU nodes in clean regions invert net lifecycle Software Carbon Intensity (SCI_{tot}), and how do decoupled host/GPU lifetimes (9 yr/3 yr vs. 4 yr) alter provisioning decisions?

This paper makes three contributions:

1) **PRISMA-2020-Guided Literature Synthesis:** Screening 102 records under explicit single-author plus AI-assisted review disclosure yields a verified 65-work corpus (60 full-text, 5 abstract-only), a dual-axis taxonomy (`L1.A–L1.F`), a direct feature comparison against the closest 2024–2026 systems, and seven open challenges (`OC1–OC7`).
2) **Benchmark-Validated Carbon-Extended Simulator:** I formalize an ISO/IEC 21031:2024 [3] lifecycle model inside SimPy 4.1.2 integrating Splitwise power/step curves [4], GreenSKU/LLMCarbon host-plus-GPU embodied parameters [28], [29], WAN KV transfer-vs-recompute gating, and GapHazard serverless control [14], [27], separated into 19 calibration anchors and 5 held-out cross-load validation checks against published baselines.
3) **Multi-Seed Trace-Driven Empirical Evaluation:** Across 10 paired seeds using CC-BY-4.0 Azure traces [30], UK NESO, and Japan TSO telemetry (with 24h trace-tiling sensitivity ablation), hybrid routing (`P4`), hazard keep-alive (`S3`), and elastic over-provisioning (`H4`) quantify the joint carbon–SLO–embodied trade-offs detailed in Section VI.

Section II reviews preliminaries; Sections III and IV present the PRISMA review, taxonomy (`RQ1`), and related work; Sections V and VI detail the simulator and multi-seed findings (`RQ1–RQ4`); Section VII discusses open challenges and limitations; and Section VIII concludes with disclosures.

II. Background and Preliminaries

Building carbon-aware LLM serving schedulers requires joining four domains that developed under separate abstractions: phase-level transformer execution, wide-area carbon routing, GPU serverless cold-start management, and lifecycle carbon accounting under ISO/IEC 21031:2024. Fig. 1 illustrates how these layers interact inside the dual-track control architecture.

A. Autoregressive LLM Phases and Disaggregation

Within cloud platforms [31], autoregressive serving separates into compute-bound *prefill* (parallel prompt processing over L_{in} tokens governing TTFT) and memory-bandwidth-bound *decode* (sequential generation over L_{out} steps reloading weights and KV state from HBM, governing TPOT). Azure traces capture sharp skew between conversational (CV = 1.094, 1,154.7 in / 211.1 out) and coding traffic (CV = 13.152, 2,047.8 in / 27.9 out) [30], while Mooncake averages 8,590 in / 182 out [32].

Local schedulers target intra-node bottlenecks first: PagedAttention cuts KV waste from 61.8%–79.6% to 3.7% (saving up to 55.2% KV memory) [33], while Sarathi-Serve eliminates 28.3× stalls via chunked prefill [6]. Phase-disaggregated engines (Splitwise [4], DistServe [5], Mooncake [32], DualScale [8]) decouple prefill and decode pools over NVLink or RDMA. Splitwise shows decode GPUs sustain 50% TDP caps (700 W to 350 W on H100) with minimal TPOT loss at 2.35× iso-power throughput [4], and DualScale saves 39% prefill and 48% decode energy at P99 parity [8]. Decode still dominates energy across token ratios [34], bilinear $(L_{\text{in}}, L_{\text{out}}, L_{\text{in}}L_{\text{out}})$ regression reaches $R^2 \geq 0.964$ [35], and Li et al. [36] caution that small-batch disaggregation can trail colocation until batch-32 KV pressure triggers colocation collapse.

B. Wide-Area Carbon Routing and KV Transfer Economics

Across 123 Electricity Maps zones, Sukprasert et al. [12] measure a 40× grid CI spread (mean 368.39 gCO_2eq/kWh), shrinking 96% unconstrained carbon gains to 51.5% at 50% spare capacity and 31% under 50 ms RTT limits, with 10%–12% error overhead at 50% forecast noise. Geo-distributed routers (GreenCourier [37] at 8.7%–17.8% carbon cuts, Wait Awhile [10], CarbonScaler [11], EcoVisor [38], Caribou [13], Calavaro et al. [39]) assume deferrable jobs or small HTTP payloads. Cross-region P/D instead ships hundreds of megabytes of 70B FP16 KV state per request, incurring serial WAN delay ($L_{\text{kv}}/B_{\text{wan}}$) and network energy (0.001–0.005 kWh/GB [13]) that can offset grid arbitrage.

C. Serverless GPU Keep-Alive and Checkpoint Loading

With 81% of Azure Functions apps receiving $\leq$ 1 req/min [40], idle nodes draw 21%–83% CPU power [14] and 60–100 W per idle GPU, whereas reloading 70B (130 GB) weights spans 84.0 s (PyTorch), 48.0 s (Safetensors), and 10.3 s (ServerlessLLM RAID-0 [27]). Adaptive controllers (EcoLife [41], CASA [42], LowCarb [43], LACE-RL at -77.08% idle carbon [14], Festina at -56% energy and -85% cold starts [44]) omit cross-grid loader sensitivity.

D. ISO/IEC 21031:2024 SCI and Embodied Carbon

To prevent shifting emissions between operational and hardware budgets, the Green Software Foundation Software Carbon Intensity (SCI) specification, standardized as ISO/IEC 21031:2024 [3], mandates an additive lifecycle intensity per functional unit R (here, 1,000 processed tokens) while explicitly excluding offset purchases:

$$\text{SCI} = \frac{(E \cdot I) + M}{R}, \tag{1}$$

where E is total consumed electricity (kWh, inclusive of datacenter PUE), I is regional grid carbon intensity (gCO_2eq/kWh), and $M = M_{\text{server}} \cdot (t_{\text{slot}}/T_{\text{life}})$ is amortized manufacturing carbon.

Industry GHG audits [2], architectural models [45], [46], and LLM carbon tools [28], [29] show embodied emissions (800 $kgCO_2eq$ host [28]; 921.6 $kgCO_2eq$ GreenSKU Genoa plus die/HBM [29]) reach 24%–35% on average grids and $> 80\%$ on clean grids. Optimizing local GPU utilization [47] or ignoring power caps [7], [48], [49] and hardware aging [50], [51] can raise lifecycle carbon, motivating unified simulators [52].

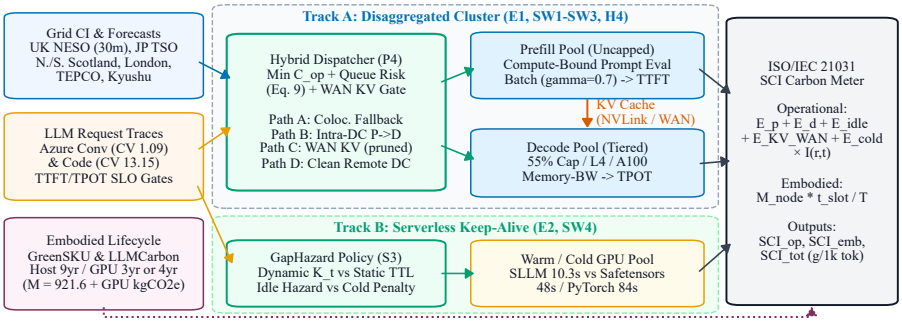


Fig. 1. Dual-track carbon-aware phase-disaggregated LLM cluster (Track A: Paths A–D via Eq. (9)) and serverless GPU keep-alive (Track B: GapHazard policy `S3`) feeding the unified ISO/IEC 21031:2024 SCI telemetry meter.

III. PRISMA-2020-Guided Review Methodology

To ground `RQ1` and parameterize the simulator from primary artifacts, I conducted a PRISMA-2020-guided literature review across Crossref, Semantic Scholar, OpenAlex, arXiv, ACM Digital Library, IEEE Xplore, USENIX, and official standards indices.

A. Search Queries, Inclusion Criteria, and Screening

Candidate studies published between 2011 and November 2026 were retrieved across three channels, expanded in revision v2 with targeted searches over 2024–November 2026 preprints and proceedings:

- **Channel 1 (Carbon-Aware Cloud & Serverless Scheduling, 16 records):** Paired carbon-intensity, grid emissions, SCI, and embodied terms with spatial/temporal shifting, serverless keep-alive, and cold-start mitigation (`Query A`).
- **Channel 2 (Phase-Disaggregated LLM Serving & v2 Update, 44 records):** Combined LLM inference, KV caching, prefill/decode disaggregation, and GPU serving with energy, DVFS, power capping, and embodied/SCI queries (`Query B`, plus 2024–2026 arXiv/Crossref follow-ups; exact boolean strings appear in the online supplement).
- **Channel 3 (Surveys, Traces, Lifecycle Models & Snowballing, 42 records):** Aggregated recent surveys, standards, hardware embodied datasets, citation chasing, 9 systems verified during revision v2 [15], [17]–[21], [26], [53], [54], and 4 additional 2025–2026 geo-distributed LLM schedulers and datacenter planners [22]–[25].

Screening Protocol: Candidate retrieval, deduplication, and full-text eligibility checks were conducted by the author with generative AI search assistance; every included entry's bibliographic metadata, DOI/arXiv identifier, and quantitative metrics were verified directly against primary PDFs, HTML preprints, or Crossref records. As shown in Fig. 2, deduplicating 102 records left 99 unique records for title/abstract screening; excluding 14 out-of-scope records (lacking cloud serving, datacenter energy, or carbon context, including 3 edge/mobile/satellite works) and 20 full-text exclusions (offline training only, latency-only disaggregation/pruning, or superseded position notes) yielded 65 included works.

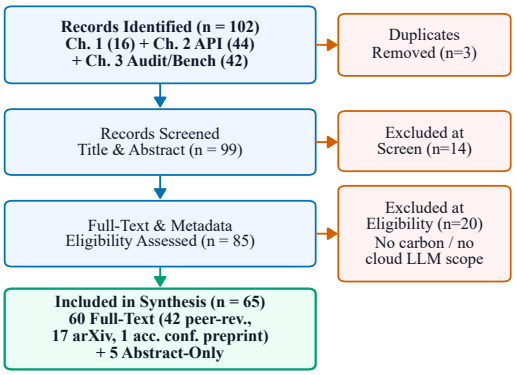


Fig. 2. PRISMA-2020-guided search, screening, and eligibility pipeline yielding the 65-work corpus (60 full-text, 5 abstract-only).

B. Corpus Provenance and Preprint Disclosure

The 65-work corpus comprises 51 empirical/modeling systems, 11 surveys or research agendas, 2 standards (ISO/IEC 21031:2024 [3] and NIST SP 800-145 [31]), and 1 trace release [30]. By recency, 60/65 works (92.3%) appeared in 2022–2026, and 38/65 (58.5%) appeared in 2025–2026. Retrieval depth breaks down as:

1) **Full-Text Audited (60/65, 92.3%):** Complete texts underwent equation and table verification, spanning 42 archival publications/standards/traces plus 18 preprints (17 arXiv-only preprints cited individually in Section IV and 1 accepted SIGOPS ATC 2026 preprint [44]).
2) **Abstract-Only Verified (5/65, 7.7%):** Five paywalled records relied strictly on verified abstracts [39], [43], [45], [47], [52], with zero constants imported into the simulator.

IV. Taxonomy and Literature Synthesis

This section answers `RQ1` by classifying the 65 retained works into six functional classes (`L1.A`–`L1.F`), mapping the 54 non-survey works across phase awareness and carbon signals, and contrasting the closest 2024–2026 systems directly against this study.

A. Dual-Axis Taxonomy (`L1.A`–`L1.F`)

1) L1.A Prior Surveys and Research Agendas ($n = 11$): Existing surveys address sustainable cloud scheduling or LLM serving separately (Table I). Cloud/serverless surveys [1], [55]–[58] omit P/D LLM serving, while LLM-efficiency

surveys [59]–[63] omit cross-region carbon arbitrage and embodied emissions. Most closely, Sisodia [26] proposes a `llm-d` Sustainable Inference Control Plane (SICP) combining operational, embodied, and intra-cluster KV terms, with zero experiments.

2) L1.B Spatial and Temporal Cloud Carbon Schedulers ($n = 15$): Single-DC temporal schedulers (Wait Awhile [10], CarbonScaler [11], EcoVisor [38], Clover [51]) and generic WAN routers (GreenCourier [37], Caribou [13], Calavaro [39], Sukprasert [12]) establish spatial carbon bounds for non-LLM tasks. Seven recent geo-distributed LLM routers schedule whole requests across regional fleets: Topcu et al. [18] (-26% to -60% CO_2 via multi-DC MILP near $\approx 3\%$ SLO violations), SLIT [19] (carbon, water, TOU across 5 sites), CarbonRouter [20] (-16.1% to -23.1% CO_2 pricing multi-turn recomputation without WAN KV migration), Bernhard and Yardimci [21] (2-DC testbed + CONUS replay, 22.7%–51.0% cuts), FREESH [22] (multi-DC LME + 1-s DVFS yielding -13.07% vs. MinEnergy; Eq. 1c writes a general P/D tuple while evaluating colocated TP pools), CEDAR [23] (CMDP+SAC across 3 AWS regions, -26% cost, -27% CO_2), and GreenLLM [24] (distinct from GreenLLM [15]; 9-DC edge LP under water caps). Across these 11 cross-region WAN schedulers (0/11 implemented), none evaluate cross-region P/D separation with WAN KV transfer gating, amortized embodied emissions, or serverless keep-alive.

3) L1.C Carbon-Aware Serverless Keep-Alive and Loaders ($n = 7$): Built on Azure Functions traces [40], CPU controllers EcoLife [41], CASA [42], LowCarb [43], and LACE-RL [14] (-51.7% cold starts, -77.08% idle carbon) adapt keep-alive windows, while GPU loaders ServerlessLLM [27] (10.3 s 70B load) and Festina [44] (-56% energy, -85% cold starts, plus -7% P/D) cut reload latency without testing keep-alive polarity across clean vs. coal grids.

4) L1.D Phase-Disaggregated and Engine-Level LLM Serving ($n = 17$): Local engines vLLM [33], Sarathi-Serve [6], Wilkins [35], ThrottLL'eM [48], SPROUT [16] ($> 40\%$ carbon cut via token LP), GreenCache [17] (5.6%–25.3% GPU vs. SSD total cut), ETCInfer [64] (33.1% energy cut via CRAC/DVFS without grid CI), DynamoLLM [7], TAPAS [49], and Fernandez [34] operate within single grids. Among the 8 disaggregated P/D designs (Splitwise [4], DistServe [5], Mooncake [32], DualScale [8], Li [36], Yan [9], Festina [44]), only GreenLLM [15] integrates grid CI and GPU lifetime reuse (-31.3% to -40.6% total carbon on one GCP cluster), while 0/8 span multiple regions, WAN KV gating, or serverless keep-alive.

5) L1.E Embodied Hardware and Lifecycle Models ($n = 12$): GHG studies [2], [3], [45], [46], LLMCarbon [29], Sinha [47], Hewage [50], EcoServe [28] (9 yr/3 yr lifetimes), Nguyen [54], HungryAI [65], Vidur+Vessim [53], and InFactPlanner [25] (H100/B200 energy within 1.9%–8.7%, abstracting away batching, KV state, P/D split, and embodied carbon) establish Scope 3 dominance without P/D control, WAN KV gating, or keep-alive.

6) L1.F Foundations and Traces ($n = 3$): NIST SP 800-145 [31], CloudSim [52], and the Azure LLM Inference Trace 2024 [30] supply taxonomy and trace baselines.

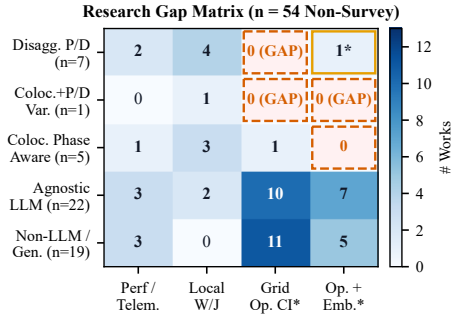


Fig. 3. Intersection of phase awareness and carbon/energy signals across $n = 54$ non-survey works. GreenLLM [15] populates P/D + Op.+Emb. (`1*`).

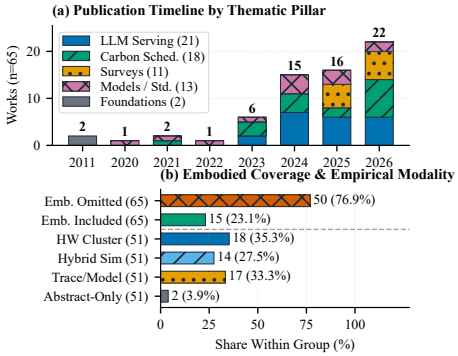


Fig. 4. Corpus bibliometrics from 2011 to Nov. 2026 ($n = 65$): (a) 2024–2026 growth (58.5% in 2025–2026); (b) 76.9% embodied omission ($n = 65$) and empirical split ($n = 51$).

B. Gap Matrix and Closest Related Work Comparison

Cross-tabulating the 54 non-survey works in Fig. 3 shows that 50/65 works (76.9%) omit embodied accounting (Fig. 4b), while the 51 empirical works split across physical clusters (18/51, 35.3%), hybrid testbed+simulators (14/51, 27.5%), trace/analytical models (17/51, 33.3%), and abstract-only records (2/51, 3.9%). Table II maps the closest 2024–2026 systems and design agendas directly against this work across all four architectural axes, and Table III summarizes the broader `L1.B`–`L1.F` classes. Across all 65 works, **zero implemented systems** jointly combine P/D disaggregation, cross-region carbon routing with explicit WAN KV transfer gating, host plus GPU embodied carbon amortization, and serverless checkpoint keep-alive.

C. Detailed Comparison Against Closest Systems

- **Topcu et al. (Penn State) [18]:** Their MILP router schedules continuous-batching LLMs across 16 US/EU DCs ($< 5\%$ TPOT, 9.1% TTFT, $< 2.5\%$ energy simulator error), cutting CO_2 by 26%–60% near $\approx 3\%$ SLO violations. Colocated A100 pools omit P/D separation, WAN KV transfer, embodied carbon, and keep-alive.
- **GreenLLM [15]:** GreenLLM disaggregates P/D with speculative decoding on A100 and reused T4/V100 GPUs within one GCP cluster, pairing grid CI with 5–7 yr GPU reuse for 31.3%–40.6% total carbon cuts at 90% SLO. Staying intra-DC, it omits WAN KV gating, host/chassis embodied share ($\geq 90\%$ clean grids), and keep-alive.
- **Complementary Works:** CarbonRouter [20] prices prompt recomputation across four US DCs without WAN KV mi-

TABLE I
AUDIT OF PRIOR SURVEYS AND RESEARCH AGENDAS (L1.A, $n = 11$)

Work	Year	Venue / Status	Primary Scope	Grid CI	Emb.	LLM	P/D & KV	Verified Gap Summary
Hewage et al. [55]	2026	arXiv (28 pp.)	Latency-sensitive cloud	✓	✓	×	×	0 LLM/P/D/KV hits; cites $> 60\%$ CO_2 cut
Yang et al. [1]	2025	arXiv (35 pp.)	Container orchestration	✓	×	Min.	×	Covers SCI, serverless, GPU; 0 P/D/KV/emb.
Alam et al. [56]	2026	*Computing* (25 pp.)	Cloud carbon RL/heur.	6.3%	×	×	×	70% (44/63) sim-only; 0 GPU/LLM/emb.
Agarwal et al. [57]	2026	*ACM CSUR* (36 pp.)	Serverless adaptation	4/40	×	Min.	×	Sustainability secondary; 0 P/D/KV/emb.
Besozzi et al. [58]	2025	*IEEE Access* (46 pp.)	Serverless arch.	2 hits	×	×	×	29.5% GPU support; 0 LLM/P/D/emb.
Miao et al. [59]	2025	*ACM CSUR* (35 pp.)	Generative LLM serving	Min.	×	✓	✓	Covers KV & disagg; 0 emb./CI/water
Bai et al. [62]	2026	*ACM CSUR* (70 pp.)	LLM serving efficiency	✓	×	✓	×	PUE·kWh·CI metrics; 0 disagg/cold/emb.
Jiang et al. [60]	2026	*ACL Findings* (27 pp.)	KV compression	×	×	✓	✓	Recommends J/tok; 0 grid CI or emb.
Pan et al. [61]	2025	arXiv (25 pp.)	Data-centric LLM sys.	×	×	✓	✓	29 prefill, 12 disagg; 0 carbon/energy
Zhen et al. [63]	2025	arXiv (20 pp.)	Efficient LLM serving	Min.	×	✓	✓	6 P/D designs; brief Sec. 6.5 energy note
Sisodia [26]	2026	arXiv (13 pp.)	`llm-d` SICP agenda	Prop.	Prop.	✓	✓	Formalizes agenda Eq. (1)/(8); 0 expts.

TABLE II
CLOSEST RELATED WORK (2024–2026) VS. THIS WORK ACROSS ALL ARCHITECTURAL AXES

System / Work	Scope	P/D Split	Multi-DC	Grid CI	WAN KV Gate	Host+GPU Emb.	Serverless KA
GreenLLM [15]	Single GCP cluster	✓(het.)	×	✓	×	GPU only	×
Topcu et al. (Penn State) [18]	16 US/EU DCs	×	✓	✓	×	×	×
Sisodia (`llm-d` SICP) [26]	Research agenda	Agenda	Agenda	Agenda	Agenda	Agenda	×
CarbonRouter [20]	4 US DCs (multi-turn)	×	✓	✓	Recompute only	×	×
Cleanest Grid [21]	2-DC live + CONUS replay	×	✓	✓(MOER)	×	×	×
SLIT [19]	5 global DCs	×	✓	✓(+water)	×	×	×
FREESH [22]	3 heterogeneous DCs	Coloc. eval.	✓	✓(LME)	×	×	×
CEDAR [23]	3 AWS regions	×	✓	✓(MOER)	×	×	×
Green-LLM [24]	9 edge DCs (+wind)	×	✓	✓(+water)	×	×	×
InFactPlanner [25]	Multi-DC planner	×	✓	✓(+solar/water)	×	×	×
GreenCache [17]	Single cluster + SSD	×	×	✓	×	SSD only	×
SPROUT [16]	Single A100 cluster	×	×	✓	×	✓(5 yr)	×
Vidur+Vessim [53]	Single-DC co-sim	×	×	✓	×	Formula only	×
EcoServe [28]	Single-DC lifecycle	×	×	✓	×	✓(9/3 yr)	×
This Work (`P4 + S3`)	**Multi-DC + Serverless Sim**	✓	✓	✓	✓	**✓(4yr & 9/3yr)**	✓

TABLE III
SYNTHESIS MATRIX ACROSS EMPIRICAL SCHEDULING, LLM SERVING, AND LIFECYCLE CATEGORIES (L1.B–L1.F)

Category / Representative Works	Class	Scope	Phase	Grid CI	Emb.	KV WAN	Keep-Alive	Primary Knobs / Metrics
Wait Awhile [10], CarbonScaler [11], EcoVisor [38], Clover [51]	L1.B	Single DC	Non-LLM	✓	×	×	×	Temporal shift, MIG, batteries
GreenCourier [37], Caribou [13], Sukprasert [12], Calavaro [39]†	L1.B	Multi-DC	Non-LLM	✓	×	×	×	Geo-routing, FaaS DAGs, LP
Topcu [18]‡, SLIT [19], CarbonRouter [20]‡, Cleanest Grid [21]‡, FREESH [22]‡, CEDAR [23], Green-LLM [24]‡	L1.B	Multi-DC	Coloc. LLM	✓	×	×	×	MILP, SAC, DVFS, MOER, LP
CASA [42], EcoLife [41], LowCarb [43]†, LACE-RL [14], ServerlessLLM [27], Festina [44]‡	L1.C	Serverl.	Mixed/P/D	Mixed	EcoLife	×	✓	Dynamic TTL, NVMe I/O, MPS
vLLM [33], Sarathi [6], DynamoLLM [7], GreenCache [17], SPROUT [16], ETCInfer [64]‡	L1.D	Engine/DC	Coloc. / Aware	Subset	Subset	×	×	DVFS, token LP, SSD cache
Splitwise [4], DistServe [5], Mooncake [32], DualScale [8]‡, Li [36]‡, Yan [9]‡	L1.D	Intra-DC	Disagg. P/D	×	×	Intra-DC	×	P/D pools, 50% cap, RDMA
GreenLLM [15]‡	L1.D	Single DC	Disagg. P/D	✓	GPU	Intra-DC	×	Het. A100+T4/V100, spec. dec.
ACT [45]†, LLMCarbon [29], EcoServe [28]‡, Nguyen [54], Vidur+Vessim [53], HungryAI [65]‡, InFactPlanner [25]‡	L1.E	Lifecycle	Agnostic	✓	Subset	×	×	SCI Eq. (1), 4R (9/3 yr), MFU
NIST [31], CloudSim [52]†, Azure Trace [30]	L1.F	Found.	Spec./Trace	×	×	×	×	Taxonomy, cloud sim, traces
This Work (`P4` Cluster + `S3` Serverless)	–	**Multi-DC**	**Disagg. P/D**	✓	✓	✓	✓	**Hybrid A–D, SCI, GapHazard**

†Abstract-only verified (5 works). ‡Preprint / accepted conference preprint (18 works across corpus).

gration; FREESH [22], CEDAR [23], and Green-LLM [24] omit WAN KV gating and embodied carbon; InFactPlanner [25] omits batching, KV state, and P/D separation; and Sisodia [26] proposes an intra-DC `llm-d` objective with zero experiments.

V. SIMULATION ARCHITECTURE AND EXPERIMENTAL DESIGN

A. Parameterized Simulation Scope Statement

All empirical results derive from a parameterized SimPy 4.1.2 discrete-event simulator without physical multi-continent GPU cluster telemetry; consistent with Alam et al. [56] (70% simulation-based cloud carbon studies), this enables controlled multi-seed ablations across heterogeneous grids. Constants split into published baselines (Table IV) and explicit calibration assumptions (Supplement Table S6), driven by CC-BY-4.0 traces.

B. Phase Execution, Power, and Dispatch Formulation

For request i arriving at time t_i with prompt length $L_{\text{in}}(i)$ and output length $L_{\text{out}}(i)$, prefill duration on GPU tier g scales linearly above base kernel overhead $\tau_{P,0}(g)$:

$$T_{\text{exec},P}(i, g) = \tau_{P,0}(g) + \tau_{P,\text{tok}}(g) \cdot L_{\text{in}}(i), \tag{2}$$

calibrated to Splitwise Table IV uncontended Llama-70B anchors ($84\,\text{ms}$ conv and $95\,\text{ms}$ code on H100; $155\,\text{ms}$ and $185\,\text{ms}$ on A100) [4]. During continuous-batching decode over

active resident KV size K_{res} (GB) on G_r tensor-parallel GPUs, step time increases with HBM bandwidth contention:

$$t_{\text{step}}(g, K_{\text{res}}) = \tau_{D,0}(g)\left(1 + \alpha_{\text{mem}}(g)\frac{K_{\text{res}}}{G_r\,\text{BW}_{\text{HBM}}(g)}\right), \quad (3)$$

anchored to 28 ms on H100 and 42 ms on A100 [4], [7], with mild step inflation only if decode TDP cap $c_{\text{cap}} < 0.50$. Under colocation (`B1`), concurrent prefill slows active decode progress to factor $r_{\text{coloc}} = 0.20$ [4], [6], whereas disaggregation isolates decode iterations. End-to-end metrics aggregate cold-start delay, queueing, execution, KV transfer, and RTT:

$$\begin{aligned}\text{TTFT}_i &= T_{\text{cold},i} + T_{\text{q},P,i} + T_{\text{exec},P,i} + T_{\text{kv},i} + \text{RTT}_i,\\ \text{TPOT}_i &= \frac{T_{D,\text{exit},i} - T_{D,\text{enter},i}}{L_{\text{out}}(i)}.\end{aligned} \quad (4)$$

Instantaneous per-GPU power scales sublinearly with Model FLOPs Utilization (mfu) and active power cap ratio $c_{\text{cap}} \in (0, 1]$ ($c_{\text{cap}} = 1.0$ on prefill; $c_{\text{cap}} = 0.55$ on decode [4]), plus GreenSKU Genoa host ancillary draw $P_{\text{host}}(\text{mfu})$:

$$\begin{aligned}P_{\text{gpu}}(\text{mfu}, c_{\text{cap}}) = P_{\text{idle}} + (c_{\text{cap}}P_{\text{TDP}} - P_{\text{idle}})\\ \times \left(\min\left(1.0, \frac{\text{mfu}}{\text{mfu}_{\text{sat}}}\right)\right)^{\gamma},\end{aligned} \quad (5)$$

using saturation threshold $\text{mfu}_{\text{sat}} = 0.45$ and calibrated sublinear exponent $\gamma = 0.70$.

For Llama-2-70B FP16 GQA-8 ($s_{\text{kv}} = 0.3125$ MB/tok), total KV payload $S_{\text{kv}}(i) = s_{\text{kv}}L_{\text{in}}(i)$ transfers over intra-DC NVLink/IB ($\delta_{\text{layer}} \in \{5, 8\}$ ms [4], $\text{BW}_{\text{local}} = 400$ Gbps) or cross-region WAN:

$$\begin{aligned}T_{\text{kv}}(r_P, r_D) &= \text{RTT}(r_P, r_D) + \frac{8S_{\text{kv}}}{B_{\text{wan}}},\\ E_{\text{kv}} &= P_{\text{net}}\frac{8S_{\text{kv}}}{3.6\times 10^6 B_{\text{wan}}}\,\text{PUE}.\end{aligned} \quad (6)$$

Cross-region pipelined disaggregation (Path C from r_P to $r_D \neq r_P$) is admissible only if both a latency recomputation crossover condition and an operational carbon crossover condition hold:

$$\begin{aligned}T_{\text{kv}}(r_P, r_D) + T_{\text{q},D}(r_D) < \min\big(T_{\text{q},D}(r_P),\\ T_{\text{recomp}}(L_{\text{in}}, r_D)\big),\end{aligned} \quad (7)$$

$$\begin{aligned}E_P I_{r_P}(t) + E_{\text{kv}}\frac{I_{r_P}(t) + I_{r_D}(t)}{2}\\ + E_D I_{r_D}(t) < \hat{C}_{\text{local}}(i).\end{aligned} \quad (8)$$

Across all feasible paths $p \in \{\text{A: Coloc., B: Intra P/D, C: Cross P/D, D: Remote Clean DC}\}$ meeting TTFT slack ($0.85 \cdot \text{TTFT}_{\text{SLO}}$), hybrid router `P4` selects:

$$\begin{aligned}p^* = \arg\min_{p\in\mathcal{F}(i,t)}\Big[w_c\hat{C}_{\text{op}}(p,t)\\ + 40\,w_q\max\big(0,\,\text{occ}_D(p) - 0.60\big)^2\Big],\end{aligned} \quad (9)$$

with $w_c = 1.0$, $w_q = 0.15$, and decode pool occupancy occ_D; when bursts exceed slack, `P4` falls back locally to Path A or Path B.

In Track B (serverless GPU serving), an online causal EWMA tracker (`OnlineGapHazardTracker`, `S3`) estimates discrete inter-arrival gap probabilities $\mathbf{p}$ across bins $[0\text{–}1, 1\text{–}5, 5\text{–}10, 10\text{–}30, 30\text{–}60, > 60]$ s and selects dynamic keep-alive timeout $k^* \in \{1, 5, 10, 30, 60\}$ s [14] at idle entry:

$$\begin{aligned}k^*(t) = \arg\min_{k\in\mathcal{K}}\Big[\mathbb{E}[\tau_{\text{idle}}(k)]\,r_{\text{idle}}(t)\\ + p_{\text{miss}}(k)\left(C_{\text{cold}}(t) + \frac{\beta_{\text{SLO}}\,\mathbb{I}_{\text{SLO}}}{\max(\widetilde{I}_r(t), 0.04)}\right)\Big],\end{aligned} \quad (10)$$

where $r_{\text{idle}}(t)$ is per-second warm-idle carbon rate ($\lambda_{\text{idle}} = 0.20$ baseline [14] or hardware idle ratio), $C_{\text{cold}}(t)$ is reload carbon during T_{cold}, and $\widetilde{I}_r(t)$ is normalized regional intensity.

Finally, operational emissions C_{op} integrate node and WAN power multiplied by PUE $= 1.12$ and regional grid CI $I_r(t)$, while amortized embodied emissions C_{emb} sum host and GPU footprints over lifetimes $\text{LT}_{\text{host}}, \text{LT}_{\text{gpu}}$ (4 yr uniform; sensitivity 9 yr host / 3 yr GPU [28]):

$$\begin{aligned}C_{\text{emb}} &= \sum_{m\in\mathcal{N}}\left(\frac{M_{\text{host},m}}{\text{LT}_{\text{host}}} + \frac{G_m M_{\text{gpu},m}}{\text{LT}_{\text{gpu}}}\right)T_{\text{sim}},\\ \text{SCI}_{\text{tot}} &= \frac{C_{\text{op}} + C_{\text{emb}}}{N_{\text{tok}}/1000}.\end{aligned} \quad (11)$$

C. Verified Baselines, Traces, and Calibrated Assumptions

Table IV lists verified hardware and loader parameters, while non-measured assumptions (RTTs 0.5–200 ms, 10–400 Gbps WAN, $\gamma = 0.70$, $f_{\text{cold}} = 0.65$, 24h tiling) appear in Supplement Table S6 and sweeps `SW1–SW6`. Cluster experiments (`E1`) span 3 zones (6 nodes, 24 H100s; 2 nodes/region): `South Scotland` (mean 9.75 gCO_2eq/kWh, 0–129), `London` (163.32, 13–326), and `TEPCO` (581.64, 302.3–671.5), from Oct. 2025 UK NESO (1,344 30-min slots) and FY2024 Japan TSO traces (17,520 slots, scaled $\times$1,000). Azure LLM conversational (`conv`: 19,366 reqs, 5.53 RPS, CV $= 1.094$, 1,154.7 in / 211.1 out) and coding (`code`: 8,819 reqs, 2.57 RPS, CV $= 13.152$, 2,047.8 in / 27.9 out) traces [30] use DynamoLLM SLOs (250/100, 400/100, 2,000/100 ms TTFT/TPOT) [7].

Cluster runs compare five routing policies: `B1` (carbon-agnostic colocated round-robin), `B2` (intra-region P/D, 55% decode cap), `P3` (unconditional cross-region P/D when RTT $\leq$ TTFT slack), `P4` (hybrid Eqs. (7) through (9)), and `P5` (oracle 30-min grid lookahead).

Serverless runs (`E2`) compare `S1` (0 s), `S2` (30 s), `S2b` (60 s), and `S3` (GapHazard Eq. (10)) across `TEPCO`, `North Scotland` (mean 35.65 gCO_2eq/kWh, std 82.50, 0–355), and `South Scotland` using ServerlessLLM's 10.3 s 70B loader. Additional robustness sweeps probe Path C activation across 10–400 Gbps WAN and 5–200 ms RTT (`SW5`, 690 runs) and clean scale-out break-even demand from 1.0$\times$ to 3.0$\times$ RPS (`SW6`, 560 runs). All suites run across 10 seeds (100–109) and report Student-t 95% CIs ($t_{0.975,9} = 2.262$) and two-sided paired Wilcoxon p-values ($\alpha = 0.05$, exact floor $p = 2/2^{10} \approx 0.00195$).

D. Simulator Calibration and Held-Out Validation

Prior to multi-region sweeps, 24 checks (`validate_simulator.py`; Supplement Table S2) separate 19 *calibration anchors* from 5 *held-out validation checks*. Across the 19 anchors calibrated to Splitwise

TABLE IV
VERIFIED HARDWARE, EMBODIED, AND LOADER PARAMETERS

Component	Verified Value(s)	Source
H100 SXM5 (4/node)	TDP 700 W, Idle 60 W, HBM 3352 GB/s, 124.9 $kgCO_2e$	[4], [28]
A100 SXM4 (4/node)	TDP 400 W, Idle 100 W, HBM 2039 GB/s, 121.5 $kgCO_2e$	[4], [28]
L4 Decode (8/node)	TDP 72 W, Idle 14.4 W, 300 GB/s, 41.2 $kgCO_2e$	[28]
GreenSKU Genoa	Peak 562.4 W, Idle 164.6 W, 921.6 $kgCO_2e$	[28], [30]
Decode Cap & KV	$c_{\text{cap}} = 0.55$; $\delta_{\text{layer}} = 5/8$ ms; PUE 1.12	[4]
70B / 13B Loaders	SLLM 10.3 / 2.0 s; Safe 48.0 / 8.2 s; PyT 84.0 / 14.0 s	[27]
Amortization Life	Uniform 4 yr; Asymmetric 9 yr host / 3 yr GPU	[2], [28]

Table IV and Figs. 8–9 [4] plus ServerlessLLM Fig. 6a [27], uncontended H100 TPOT (0.00% `conv`, -9.68% `code`), loader times (0.00% across 2.0–84.0 s), and 50%–100% TDP step stability (4.47% step delta; +6.0% below 50%) match by construction; as an explicit modeling limitation, modeled TTFT deviates by **13.73%–14.24%** on H100 (95.96/108.04 vs. 84.0/95.0 ms) and **8.81%–9.50%** on A100 (169.73/201.29 vs. 155.0/185.0 ms) due to fixed 5–8 ms transfer stalls. Across the 5 held-out `conv` checks (0.25x–2.00x RPS), unfitted runs reproduce Li et al. [36]: `B2` vs. `B1` energy drops from -20.01% (0.25x) to -30.40% (1.50x) as P95 TTFT rises +11.4% to +44.3% (2.00x) and P95 TPOT improves -7.8% to -31.6%.

E. 24-Hour Trace-Tiling Sensitivity Ablation

Because raw Azure traces span 58.36 min (`conv`) and 57.27 min (`code`), I tested 24h block-tiling sensitivity across four modes ($N = 10$ `conv` seeds; `run_tiling_ablation.py`, full 12-row results in Supplement Table S3). Relative `P4` SCI_{op} savings stay invariant within 0.12 pp (60.70%–60.82% vs. `B1`; 44.07%–44.33% vs. `B2`), and SLO violations hold within 0.817%–0.835% across original 280 s replacement, 280 s no-replacement permutation, fine 90 s, and coarse 900 s regimes.

VI. SIMULATION RESULTS AND DISCUSSION

All inferential comparisons span $N = 10$ paired seeds (100–109). Because $p = 2/2^{10} \approx 0.00195$ is the minimum two-sided exact paired Wilcoxon signed-rank p-value for $N = 10$ when all pairs share sign ($|r_{\text{rb}}| = 1.000$), I report paired mean differences (Δ), Student-t 95% CIs, r_{rb}, paired Cohen's d_z, and exact p-values (0.002 at three decimals; full statistics in the online supplement).

A. E1: Multi-Region Cluster Carbon–SLO Frontier

Table V and Fig. 5 summarize `E1` across 1h and 24h horizons. On conversational traffic (`conv`), hybrid routing `P4` lowers SCI_{op} by 60.77% at 1h ($\Delta = -0.05124$ [-0.05473, -0.04775] g/1k tok, $r_{\text{rb}} = -1.000$, $d_z = -10.50$, $p = 0.002$) and 60.70% at 24h ($\Delta = -0.04948$ [-0.05220, -0.04676], $d_z = -13.01$, $p = 0.002$) relative to carbon-agnostic colocation `B1`, and by 43.95%–44.07% relative to intra-region phase disaggregation `B2` ($d_z \in [-19.99, -11.29]$, $p = 0.002$). Because fixed hardware amortization adds $SCI_{emb} \approx 0.0090$ gCO_2eq/1k tok across equal-capacity clusters, total SCI_{tot} drops 54.92% (1h) and 54.64% (24h) versus `B1` at 0.81%–0.82% SLO violations (TTFT P95 = 504.5–506.4 ms), well below the 5% boundary.

By contrast, unconditional spatial disaggregation `P3` reaches lower SCI_{tot} (0.03858 at 24h) at the cost of tail degradation, breaching 5% at 24h (5.90% violations, TTFT P95 = 1,190.4 ms). `P4` cuts `P3` SLO violations by 83.57% (1h, $\Delta = -4.12$ pp [-5.12, -3.11]) and 86.04% (24h, $\Delta = -5.08$ pp [-6.16, -4.00], $r_{\text{rb}} = -1.000$, $p = 0.002$) for a +6.48%–6.79% SCI_{tot} trade-off, supporting **H1** and **H2**.

Under bursty coding completions (`code`, CV = 13.152, mean 2,047.8-token prompts; Fig. 5b), `P4` cuts SCI_{op} by 28.51% versus `B1` ($\Delta = -0.01763$ [1h], -0.01685 [24h], $d_z \in [-13.14, -9.73]$, $p = 0.002$) and 23.45%–23.51% versus `B2` ($p = 0.002$). Unconditional `P3` saturates the WAN with 192.5–193.1 GB/1k-req KV transfers, driving SLO violations to 44.45%–44.57% (TTFT P95 = 30.7–39.3 s). To trace where spatial gains originate, `P4`'s gates prune cross-region Path C cleanly to **0.00%** at 10–25 Gbps (Section VI-C). Consequently, `P4`'s operational carbon reductions come primarily from **Path D** (whole-request remote clean-grid execution: 66.13%–66.27% on `conv`, 46.08%–46.74% on `code`) paired with local intra-DC fallbacks onto Path B (33.54%–39.81%) and Path A (0.20%–14.11%), cutting `P3`'s `code` SLO violations by 85.22% (1h, $\Delta = -37.88$ pp [-38.21, -37.55], $d_z = -82.83$, $p = 0.002$) and 86.01% (24h, $\Delta = -38.34$ pp [-38.57, -38.10], $d_z = -118.80$). Oracle 30-min grid lookahead `P5` matches `P4` within 0.02% on carbon and 0.25% on SLOs ($p \geq 0.160$).

B. E2 & SW4: Serverless Keep-Alive and Loader Sensitivity

Table VI and Fig. 6 reveal an inverse regional trade-off for dynamic keep-alive (`S3`). On carbon-intensive `TEPCO` (581.6 gCO_2eq/kWh), 24h GapHazard `S3` achieves the lowest operational intensity ($SCI_{op} = 0.10825$), trimming operational carbon by 8.05% below scale-to-zero `S1` ($\Delta = -0.00948$ [-0.01058, -0.00838], $r_{\text{rb}} = -1.000$), 7.03% below 30 s static `S2` ($\Delta = -0.00819$ [-0.00873, -0.00764]), and 11.02% below 60 s `S2b` ($\Delta = -0.01340$, all $p = 0.002$). However, aggressive dirty-slot eviction leaves 28.60% SLO violations (4,529 cold starts) versus 19.45% (`S2`) and 13.03% (`S2b`), though well below `S1` (40.25%), partially supporting **H3**.

Conversely, on clean `South Scotland` (9.75 gCO_2eq/kWh), inverse intensity scaling in Eq. (10) extends warm retention during clean intervals, cutting cold-start SLO violations by **32.63%** relative to 30 s `S2` ($\Delta = -6.29$ pp [-6.60, -5.99], 12.99% vs. 19.29%, $r_{\text{rb}} = -1.000$, $p = 0.002$) while statistically tying 60 s `S2b` (12.84%, $p = 0.125$) at numerically lower SCI_{op} (0.00094, $p = 0.084$ vs. `S1`/`S2`). Embodied hardware accounts for 97.4%–97.6% of clean-grid lifecycle SCI ($SCI_{emb} = 0.03308$), supporting **H4**. Volatile `North Scotland` displays zero-carbon wind slots that erase operational differences ($p \geq 0.50$) while retaining the 30.50% SLO reduction versus `S2` ($\Delta = -5.88$ pp, $p = 0.002$).

In `SW4` (1h `TEPCO`, Fig. 6c), slow 70B loaders amplify `S3` operational carbon savings: 23.74% vs. `S1` under PyTorch (0.46686 → 0.35601 at 84 s) and 30.93% under Safetensors (0.35452 → 0.24488 at 48 s), whereas fast `opt_13b` (2.0 s)

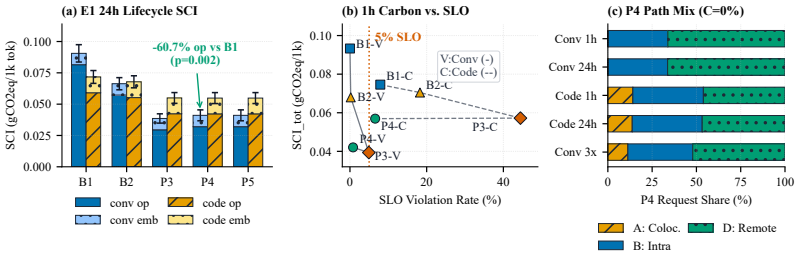


Fig. 5. Experiment `E1` cluster evaluation ($N = 10$ seeds, 95% Student-t CIs): (a) 24h operational plus embodied SCI; (b) 1h carbon–SLO frontier against the 5% boundary; (c) adaptive `P4` path mix showing Path C pruned to 0% at 10 Gbps WAN.

TABLE V
EXPERIMENT `E1` MULTI-REGION CLUSTER METRICS (MEAN ± 95% CI, $N = 10$ SEEDS; PAIRED VS. `P4`)

Case	Policy	SCI_{op} (g/1k)	SCI_{tot} (g/1k)	TTFT P95 (ms)	SLO (%)	Effect (r_{rb}, p)	Verdict (H1/H2)
1h Conv	`B1` (Coloc.)	0.08431 ± 0.00681	0.09329 ± 0.00681	268.6 ± 0.0	0.0005 ± 0.001	$-1.00, 0.002$ (op)	Baseline
1h Conv	`B2` (Intra P/D)	0.05901 ± 0.00472	0.06799 ± 0.00472	300.3 ± 0.2	0.20 ± 0.04	$-1.00, 0.002$ (op)	Baseline
1h Conv	`P3` (Uncond.)	0.03040 ± 0.00306	0.03938 ± 0.00306	1062.3 ± 136.1	4.93 ± 1.02	$-1.00, 0.002$ (SLO)	Tail Degrad.
1h Conv	`P4` (Hybrid)	$\mathbf{0.03307 \pm 0.00365}$	$\mathbf{0.04205 \pm 0.00365}$	$\mathbf{506.4 \pm 2.1}$	$\mathbf{0.81 \pm 0.06}$	–	**SUPPORTED**
24h Conv	`B1` / `B2`	0.08150 / 0.05726	0.09055 / 0.06631	268.6 / 300.3	0.00 / 0.20	$-1.00, 0.002$ (op)	Baselines
24h Conv	`P3` (Uncond.)	0.02953 ± 0.00369	0.03858 ± 0.00371	1190.4 ± 156.0	5.90 ± 1.10	$-1.00, 0.002$ (SLO)	Breaches 5%
24h Conv	`P4` (Hybrid)	$\mathbf{0.03203 \pm 0.00434}$	$\mathbf{0.04108 \pm 0.00436}$	$\mathbf{504.5 \pm 3.7}$	$\mathbf{0.82 \pm 0.03}$	–	**SUPPORTED**
1h Code	`B1` / `B2`	0.06182 / 0.05773	0.07457 / 0.07048	592.8 / 3209.1	7.91 / 18.25	$-1.00, 0.002$ (op)	CV 13.15
1h Code	`P3` (Uncond.)	0.04441 ± 0.00385	0.05722 ± 0.00385	39317.8 ± 793.9	44.45 ± 0.28	$-1.00, 0.002$ (SLO)	WAN Collapse
1h Code	`P4` (Hybrid)	$\mathbf{0.04419 \pm 0.00388}$	$\mathbf{0.05694 \pm 0.00388}$	$\mathbf{1613.8 \pm 4.3}$	$\mathbf{6.57 \pm 0.25}$	–	**H1 Part./H2 Sup.**
24h Code	`B1`/`B2`/`P3`	0.05911/0.05525/0.04235	0.07171/0.06784/0.05495	612.5/3380/30725	8.46/19.47/44.57	$\leq -0.82, \leq 0.020$	Baselines
24h Code	`P4` / `P5`	**0.04226** / 0.04226	**0.05485** / 0.05485	**1598.9** / 1599.0	**6.24** / 6.23	$\lvert r_{rb}\rvert \leq 0.20, \geq 0.652$	**H1 Part./H2 Sup.**

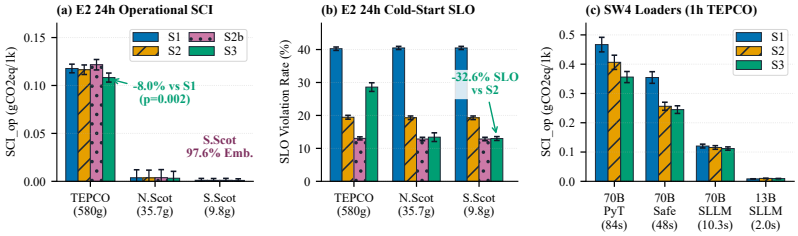


Fig. 6. Experiment `E2` and `SW4` serverless evaluation ($N = 10$ seeds, 95% CIs): (a) 24h operational SCI; (b) 24h cold-start SLO violations across grid intensities; (c) `SW4` loader sensitivity (84 s PyTorch down to 10.3 s ServerlessLLM and 2.0 s 13B).

TABLE VI
EXPERIMENT `E2` 24H SERVERLESS KEEP-ALIVE METRICS ($N = 10$ SEEDS)

Region	Pol.	SCI_{op}	Emb. (%)	SLO (%)	Cold Starts
`TEPCO`	`S1` (0 s)	0.11773	22.2	40.25	6,271.6
`TEPCO`	`S2` (30 s)	0.11644	22.4	19.45	3,054.7
`TEPCO`	`S2b` (60 s)	0.12166	21.6	**13.03**	2,081.0
`TEPCO`	`S3` (GapHaz.)	**0.10825**	23.7	28.60	4,529.0
`N. Scot.`	`S1` (0 s)	0.00373	94.6	40.48	6,373.4
`N. Scot.`	`S2` (30 s)	0.00362	94.7	19.29	3,069.1
`N. Scot.`	`S2b` (60 s)	0.00378	94.6	**12.84**	2,073.4
`N. Scot.`	`S3` (GapHaz.)	**0.00323**	**95.0**	13.40	2,164.8
`S. Scot.`	`S1` (0 s)	0.00106	97.4	40.48	6,373.4
`S. Scot.`	`S2` (30 s)	0.00103	97.4	19.29	3,069.1
`S. Scot.`	`S2b` (60 s)	0.00108	97.4	**12.84**	2,073.4
`S. Scot.`	`S3` (GapHaz.)	**0.00094**	**97.6**	12.99	2,098.4

favors `S1` on carbon (0.00831 vs. 0.00936) at 16.02% vs. 3.60% SLO violations. Replacing the 20% idle ratio with H100's 8.6% hardware idle ratio (60 W/700 W) cuts `S3` SCI_{op} by another 14.43% (0.11193 → 0.09578).

C. Sensitivity Sweeps (`SW1`–`SW6`) and Clean Scale-Out Break-Even (`H4`)

Fig. 7, Fig. 8, and Supplement Table S7 expose regime boundaries across load, WAN bandwidth/RTT, hardware tiering, and cluster capacity:

1) **SW1 Load Scaling (0.5x–3.0x conv):** At 3.0x load, `P3` saturates clean pools (71.94% SLO viol., TTFT P95 = 446.7 s), `B1` stalls via colocation (54.48% viol., TPOT P95 = 2,331 ms), and `B2` reaches 20.00%. Activating 11.30% Path A local spillover with Path B (36.62%) and Path D (52.08%), `P4` holds 2.89% violations ($SCI_{op} = 0.03483$ vs. `B1` 0.04943).
2) **SW2 Grid & Forecast Noise:** UK NESO forecasts, 30–60 min lags, and 2× noise shift `P4` SCI_{tot} by $< 0.3\%$. Across a low-differential UK pair (`South Scotland` + `North West England`), `P4` holds 0.08% violations ($SCI_{op} = 0.00513$) vs. `P3`'s 10.71% (0.00425).
3) **SW3 & SW5 Hardware Tiers and Path C Regime:** Under `L4` decode nodes, `B2` (24.73%) and `P3` (53.47%) suffer decode SLO collapse, whereas `P4` holds 0.76% violations (Fig. 7b). Supplement Table S7 and the 690-run `SW5` sweep (Fig. 8, full matrix in Supplement Table S4) explain WAN Path C pruning: raising WAN bandwidth from 10 to 100 Gbps halves `P3`'s `code` violations from 44.45% to 20.78%, whereas under `P4` at 10–25 Gbps and ≥ 20 ms RTT, Path C stays exactly **0.000%**. Opening 100–400 Gbps links at 5–50 ms RTT under 2.0× `conv` congestion activates

Path C up to **0.308%** (high ΔCI, 400 Gbps/5 ms) and **0.228%** (low ΔCI), vs. at most 0.060% at $1.0\times$ nominal load and $\leq 0.009\%$ on $2.0\times$ `code`; Path D whole-request routing (38.9%–63.0%) dominates cross-region KV transfers whenever both phases fit clean capacity. Under a strict 125 ms TTFT cap (`strict_code`), every fixed-capacity policy exceeds 5% (`P4` 48.69%, `B1` 53.30%).

4) **H4 & SW6 Clean Scale-Out Break-Even:** Across 24h `E1`, clean `South Scotland` reaches 93.0% embodied share on `code` (90.7% on `conv`). At $1.0\times$ UK demand, adding four spare clean nodes (`P3_overprov_plus4` / `P4+4`, 10 vs. 6 nodes) trims SCI_{op} by only 3.94% (0.00804 vs. 0.00837) while raising lifecycle SCI_{tot} by **+32.61%–32.66%** under 4-year uniform life (0.02300–0.02301 vs. 0.01735) and **+27.71%–27.76%** under 9 yr host / 3 yr GPU lifetimes (0.01937 vs. 0.01516), supporting **H4**. Across the 560-run `SW6` load sweep (Fig. 7c, Supplement Table S5), this penalty shrinks at $1.25\times$ (+26.96% / +21.56%) and $1.50\times$ (+13.62% / +7.30%), crosses a capacity break-even near **1.6–1.7**$\times$ RPS once the 6-node clean pool spills into dirtier UK zones, and flips carbon-negative from $1.75\times$ (-2.59% / -9.51%) to $3.0\times$ ($-\mathbf{30.63}\%$ 4-yr; $-\mathbf{36.45}\%$ 9/3-yr, cutting SLO violations from 1.17% to 0.20%).

D. Summary Answers to Research Questions (RQ1–RQ4)

- **RQ1:** Single-cluster P/D [15] and colocated multi-DC routers [18]–[24] exist, yet 50/65 works (76.9%) omit embodied carbon, 0/11 geo-routers model P/D WAN KV transfer, and 0 unify all four axes with keep-alive.
- **RQ2:** Hybrid routing (`P4`) cuts `conv` SCI_{op} by 60.70%–60.77% (SCI_{tot} 54.64%–54.92%, $|r_{\text{rb}}| = 1.000$) at $< 1\%$ SLO violation and trims `P3 code` SLO violations by 85.22%–86.01% by holding Path C at 0.000% (≤ 25 Gbps, ≥ 20 ms RTT) and $\leq 0.308\%$ on congested 400 Gbps links.
- **RQ3:** GapHazard (`S3`) cuts `TEPCO` SCI_{op} by 7.03%–11.02% (23.74%–30.93% at 48–84 s loaders; 28.60% SLO viol.) and trims clean `South Scotland` SLO violations by 32.63% vs. 30 s TTL.
- **RQ4:** Embodied carbon hits 93.0%–97.6% on clean grids; $+4$ clean nodes raise net SCI_{tot} by +32.61%–32.66% (4-yr) and +27.71%–27.76% (9/3-yr) at $1.0\times$ load before turning carbon-negative above 1.6–1.7$\times$ RPS.

VII. Open Challenges and Limitations

A. Synthesized Open Research Challenges (OC1–OC7)

Synthesizing the 65-work corpus (Section IV) and Section VI identifies seven challenges:

- **OC1 (Cross-Region KV Transport):** Since Eqs. (7)–(8) prune Path C below 0.009% (`code`) and $\leq 0.308\%$ even on congested 400 Gbps links, wide-area P/D splitting needs token KV compression [60] under live CI gates.
- **OC2 (Marginal vs. Average CI):** Multi-megawatt bursts require marginal and curtailment signals beyond the average CI used in `E1`.
- **OC3 (Demand-Indexed Embodied Sizing):** Because $+4$ clean nodes flip from +27.76%–32.66% net SCI_{tot} at $1.0\times$ load to -30.63%–36.45% at $3.0\times$ past 1.6–1.7$\times$ RPS (`SW6`), sizing must combine routing with demand-gated sleep and GPU reuse [15], [28].
- **OC4 (Serverless Checkpoint Sharing):** Closing the 8.05% vs. 28.60% `TEPCO` carbon–SLO trade-off requires memory-deduplicated GPU weight pools across adapters [44].
- **OC5 (Water–Energy–Carbon Co-Design):** Following SLIT [19] and CASA [42], routing toward dry renewable grids shifts cooling water stress unless water scarcity enters Eq. (1).
- **OC6 (Bursty Code Admission Control):** Under CV $=$ 13.152, fixed 6-node pools exceed 5% SLO violations, motivating elastic spot failover or prefill-priority admission.
- **OC7 (Multi-Continent Hardware Testbeds):** GreenLLM [15] evaluates single-cluster P/D and Bernhard and Yardimci [21] steer two colocated sites, leaving multi-continent P/D, WAN KV serialization, embodied accounting, and keep-alive untested together on hardware.

B. Limitations and Threats to Validity

Six methodological boundaries govern the interpretation of this study:

1) **Trace-Driven Simulation Abstraction:** Outcomes derive from SimPy 4.1.2 execution rather than multi-continent GPU clusters; unmodeled RDMA tail jitter, PCIe thermals, and fragmentation may shift absolute latencies, although relative bottlenecks match held-out baselines (Section V-D).
2) **Calibrated Non-Measured Constants:** Synthetic WAN RTTs (20–200 ms), bandwidths (10–100 Gbps), MFU exponent $\gamma = 0.70$, and cold power fraction $f_{\text{cold}} = 0.65$ are literature-bounded assumptions (Supplement Table S6) swept across `SW3`–`SW6`.
3) **24-Hour Trace Tiling:** Raw Azure traces span 58.36 min (`conv`) and 57.27 min (`code`) [30]; Section V-E tests sensitivity across block lengths and replacement regimes.
4) **Static Dispatch vs. Dynamic Server Power-Off:** Inside Eq. (9), per-request `P4` scores operational carbon plus queue pressure; embodied amortization compares fixed versus over-provisioned cluster topologies (`H4`) rather than dynamically powering nodes down mid-run.
5) **Corpus Access and Preprints:** Five works (7.7%) were verified solely from abstracts [39], [43], [45], [47], [52], and 18 full-text entries (27.7%) are preprints (17 arXiv-only plus 1 accepted ATC 2026 preprint [44]) subject to peer-review revision.
6) **Search-Strategy Dependence and Screening Protocol:** Retained corpus counts depend on index cutoffs (November 2026) and query phrasing (Channels 1–3, listed in the online supplement); single-author AI-assisted screening was mitigated by inspecting every work against primary full texts and Crossref DOIs.

VIII. Conclusion

This work examined carbon-aware cloud/serverless scheduling and phase-disaggregated LLM serving via a PRISMA-2020-guided review of 65 studies (60 full-text, 5 abstract-only) and a

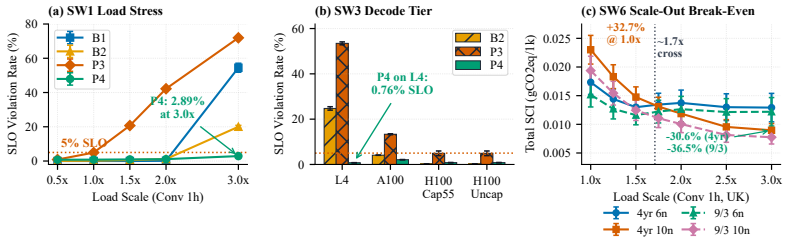


Fig. 7. Robustness and lifecycle evaluations ($N = 10$ seeds, 95% CIs): (a) `SW1` load scaling (`P4` holds 2.89% at 3.0x); (b) `SW3` decode GPU tiers (`P4` holds 0.76% on `L4`); (c) `SW6` scale-out break-even curve crossing from $+32.66\%$ (4-yr) / $+27.76\%$ (9/3-yr) penalty at $1.0\times$ to -30.63% / -36.45% net SCI savings at $3.0\times$.

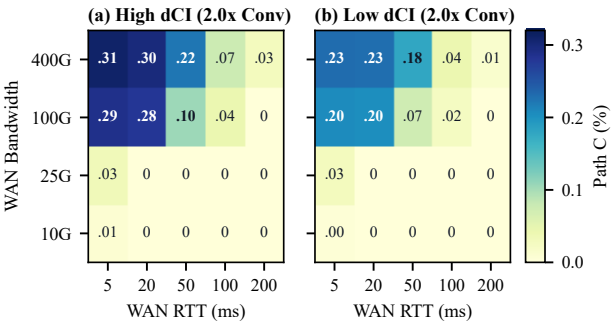


Fig. 8. Cross-region Path C activation (%) across WAN bandwidth and RTT (`SW5`, $2.0\times$ `conv`, $N = 10$ seeds): (a) high ΔCI (`South Scotland-TEPCO`); (b) low ΔCI (`South Scotland-NW England`).

SimPy 4.1.2 simulator validated against published benchmarks using Azure, UK NESO, and Japan TSO traces. While prior systems tackle single-cluster P/D carbon/GPU reuse [15] or colocated multi-DC routing [18]–[24], 76.9% of reviewed studies omit manufacturing carbon, 0/11 implemented cross-region carbon schedulers evaluate P/D WAN KV transport, and none jointly combine all four control axes with serverless keep-alive.

Trace-driven multi-seed simulations show that unconditional spatial disaggregation (`P3`) suffers tail collapse under bursty coding traffic (44.45%–44.57% SLO violations) and $3\times$ load spikes (71.94%). Hybrid carbon-and-queue routing (`P4`) preserves 0.81%–0.82% SLO violations on conversational workloads while cutting operational SCI by 60.70%–60.77% and lifecycle SCI by 54.64%–54.92% relative to colocation ($|r_{\text{rb}}| = 1.000$, exact $p = 2/2^{10} \approx 0.00195$, reported at three decimals as 0.002), and lowers `code` SLO violations by 85.22%–86.01% below `P3` by pruning cross-region Path C (0.000% at ≤ 25 Gbps / ≥ 20 ms RTT, and $\leq 0.308\%$ even on congested 400 Gbps / 5 ms links) in favor of Path D remote clean-grid execution and local Path A/B fallbacks. In serverless deployments, dynamic GapHazard keep-alive (`S3`) trims dirty-grid `TEPCO` operational SCI by 7.03%–11.02% (amplified to 23.74%–30.93% for slow 48–84 s loaders) while curbing clean-grid `South Scotland` SLO violations by 32.63% below static 30 s timeouts. Finally, hardware amortization constitutes 93.0%–97.6% of clean-grid lifecycle emissions, creating a utilization-dependent sizing rule: four spare clean nodes raise net total SCI by +27.71% to +32.66% at $1.0\times$ nominal demand, cross break-even near 1.6–$1.7\times$ RPS, and cut lifecycle SCI by 30.63% to 36.45% at $3.0\times$ demand. Future physical testbeds should pair power-capped intra-region disaggregation with WAN KV compression, demand-indexed clean capacity sizing, and shared NVMe checkpoint pools.

Data and Code Availability

External datasets (Azure LLM Inference Traces and GreenSKU tables, UK NESO Carbon Intensity API, and Japan FY2024 TSO factors) are publicly available; simulator code, PRISMA records, and reproduction scripts are hosted in a repository omitted here (*[anonymized for double-blind review]*).

Acknowledgments

During the preparation of this manuscript, generative AI language and coding assistants were used for literature search assistance (Section III), SimPy simulator and figure scripting assistance (Sections V and VI), drafting, and language editing. Every reference, quantitative claim, simulation output, and figure was reviewed and verified directly against primary sources and raw CSV outputs by the author, who takes full responsibility for the published content. Author identities, affiliations, and repository links are omitted here for double-blind review.

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